

PART VII

Physical Activity for Health and Fitness

In previous parts of this book, we focused on the physiological bases of physical activity and sport performance, describing the physiological responses to an acute bout of exercise, the adaptations to chronic training, and the means of improving performance in sport. In [part VII](#), we shift our focus from athletic performance to a special area of exercise physiology: the role of physical activity in improving and maintaining overall physical fitness and health. In [chapter 20](#), Prescription of Exercise for Health and Fitness, we discuss how to design an exercise program that can improve health-based physical fitness. We consider the essential components, ways to tailor the program to an individual's specific needs, and the unique role exercise plays in the rehabilitation of people with illness or disease. In [chapter 21](#), Cardiovascular Disease and Physical Activity, we examine the major types of cardiovascular disease, their physiological bases, and how physical activity can help prevent or slow the progression of these diseases. Finally, in [chapter 22](#), Obesity, Diabetes, and Physical Activity, we examine some of the causes of obesity and diabetes, the health risks associated with each, and the ways in which physical activity can be used to control both disorders.





20

Prescription of Exercise for Health and Fitness

In this chapter and in the web study guide

Health Benefits of Regular Physical Activity and Exercise

Physical Activity Recommendations

VIDEO 20.1 presents Bob Sallis with a physician's perspective on the prescription of exercise.

Health Screening

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AUDIO FOR TABLE 20.1 explains preparticipation health screening guidelines.

VIDEO 20.2 presents Bob Sallis on whether a physician should be consulted before beginning moderate exercise.

ACTIVITY 20.1 *Who Needs Medical Clearance?* reviews new guidelines for assessing whether a person needs clearance before starting an exercise program.

ACTIVITY 20.2 *The Exercise Electrocardiogram* reviews terms related to exercise ECGs and explains the factors that affect the accuracy of this test's results.

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ACTIVITY 20.3 *Monitoring Exercise Intensity* explores methods of monitoring exercise intensity (your own and someone else's).

Exercise Programming

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VIDEO 20.3 presents Malachy McHugh on the timing and duration of stretching, and the role of stretching in injury prevention.

Exercise and Rehabilitation of People with Diseases

In Closing

Statistics about the average adult's health and physical activity patterns do not paint a bright picture. The President's Council on Physical Fitness¹⁵ lists the following:

- Only one in three children are physically active every day.
- Less than 5% of adults participate in 30 min of physical activity each day.
- Only 1 in 3 adults receive the recommended amount of physical activity each week.
- 80.2 million people above the age of 6 are physically inactive.
- Only 35% to 44% of adults 75 years or older and 28% to 34% of adults ages 65 to 74 are physically active.
- More than 80% of adolescents and adults do not meet the guidelines for both aerobic and muscle-strengthening activities.
- Children now spend more than 7.5 h/day in front of a TV, video game, or computer screen.
- Nearly one-third of high school students play video or computer games for ≥3 h on an average school day.

Although the human body is designed for movement, modern lifestyles have channeled the average American into an increasingly sedentary existence. Decades of research have clearly and unequivocally determined that an active lifestyle is important for optimal health. From a health perspective, we have not adapted well to this inactive lifestyle.

Health Benefits of Regular Physical Activity and Exercise

It seems rather ironic that it took research from clinicians and scientists to reach the conclusion that Galen identified in the first century that regular exercise was as important as breathing fresh air and eating the proper foods in promoting health (see the Introduction chapter)!

The first acknowledgment from the modern medical profession came in 1992, when the American Heart Association proclaimed physical inactivity a major risk factor for coronary artery disease (CAD), placing it alongside smoking, abnormal blood lipids, and hypertension. Subsequently, the Centers for Disease Control and Prevention (CDC), in collaboration with the American College of Sports Medicine (ACSM), stated the importance of physical activity as a public health initiative and then in 1995 published a consensus statement by a panel of the National Institutes of Health (National Heart, Lung, and Blood Institute) advocating physical activity as important for cardiovascular health. Finally, in 1996, the Surgeon General of the United States released a written report on the health benefits of physical activity. This was a landmark report recognizing the importance of physical activity in reducing the risk for chronic degenerative diseases.

Much of the research supporting the benefits of physical activity in reducing the risk of developing chronic degenerative disease has come from the field of epidemiology, in which large populations are studied and associations between activity levels and disease risk determined. In the year 2000, exercise physiologists joined the war on what they termed the “sedentary death syndrome” by forming an action group advocating governmental support for research into the diseases and disorders associated with a sedentary lifestyle.³

Whether things have gotten better since that time is up for debate. Fewer adults now smoke cigarettes and fewer adolescents use alcohol or illicit drugs. And more adults are meeting physical activity guidelines, although we have a long way to go. Regular physical activity includes participation in moderate and vigorous physical activities and muscle-strengthening activities. Only one in three adults receives the recommended amount of physical activity each week, and more than 80 million Americans above the age of 6 are physically inactive.¹⁵

The health benefits associated with regular physical activity are undeniable, and more and more research is documenting these benefits.¹ There are well-documented effects of exercise on premature death, as well as cardiovascular disease, hypertension, and stroke. Other diseases that are minimized by regular physical

activity include type 2 diabetes, osteoporosis, obesity and metabolic syndrome, many types of cancers, and depression. Further, physical and cognitive function are both improved by a program of regular exercise. These inverse associations between physical activity and disease and disability follow dose–response relations, i.e., more physical activity is associated with large impacts on the adverse health condition.

Physical Activity Recommendations

Despite these somewhat discouraging statistics, most Americans are aware that exercise is an integral part of preventive medicine. Yet people often equate exercise with jogging 8 km (5 mi) a day or lifting weights until their muscles can do no more. Many believe that high volume and intensity of exercise training are necessary to attain health-related benefits, yet this is not true. This myth was the primary focus of the CDC/ACSM report published in 1995, which concluded that significant health benefits can be obtained if one includes a moderate amount of physical activity, such as 30 min of brisk walking, 15 min of running, or 45 min of playing volleyball, on most, if not all, days of the week. The major emphasis of this report was that through a modest increase in daily activity, most people can improve their health and quality of life. In fact, a study of older adults (70-82 years) showed that just being more active greatly reduced the risk of mortality, independent of a formal exercise program.¹⁰

RESEARCH PERSPECTIVE 20.1

Sitting, Physical Activity, and Mortality

The association of sedentary behavior (i.e., sitting) with chronic disease and mortality outcomes is now apparent and has significant clinical implications. Using objectively collected accelerometry data, the U.S. National Health and Nutrition Examination Survey (NHANES) estimates that children and adults spend nearly 8 h per day being sedentary. A common question is whether the association between sedentary behavior and chronic disease is modulated by physical activity. A recent meta-analysis revealed that the deleterious impact of sitting on all-cause mortality was attenuated at increasingly higher levels of physical activity. In other words, participating in 60 to 75 min of moderate- to vigorous-intensity activity per day essentially eliminated the chronic disease risk attributed to sitting.

An important follow-up question then becomes whether the association between moderate- to vigorous-intensity physical activity and chronic disease and mortality is modulated by sedentary behavior. Researchers recently attempted to answer this question by re-analyzing the results of the meta-analysis mentioned previously.⁸ Their results demonstrate a dose–response relation between moderate- to vigorous-intensity physical activity and mortality across all levels of sitting. In other words, even in individuals with little time spent sitting per day, more physical activity still reduces mortality risk. Thus, regardless of the amount of time spent sitting in one day, public health policy must focus on *both* reducing sedentary behavior and increasing moderate- to vigorous-intensity physical activity in order to maximize health benefits and reduce mortality risk.

This recommendation to simultaneously reduce sitting and increase physical activity is often referred to as the displacement hypothesis (i.e., displace activities with prolonged sitting with exercise or nonexercise physical activities). Implicit in this hypothesis is that replacing sitting time with physical activity will be associated with lower disease risk; however, this has not been experimentally evaluated. Another group of investigators¹¹ estimated the mortality benefits associated with the replacement of sitting time with different types of physical activity, including exercise and sports and nonexercise activity (e.g., household chores, daily walking). In this study, greater overall time spent sitting (≥ 12 versus < 5 h/day) was associated with a graded increase in risk of all-cause mortality as well as greater risk for cardiovascular mortality. In less active adults, replacement of 1 h/day of sitting with an equal amount of either exercise or nonexercise activity was associated with lower mortality. Interestingly, for adults who engage in more overall activity, the majority of which is nonexercise, replacing sitting time with even more nonexercise activity does not confer additional benefits; instead, purposeful and planned exercise is required to further reduce risk for mortality in active adults. Taken together, the results of this study provide strong support for the public health recommendation to replace inactivity with more physical activity. This displacement is associated with greater longevity, particularly so for less-active adults.

Additional health benefits can be gained through greater amounts of physical activity. Research suggests that people who can maintain a regular regimen of activity that is longer in duration or more vigorous in intensity are likely to derive greater benefit. It is now apparent that the appropriate exercise type and intensity vary depending on individual characteristics, current fitness level, and specific health concerns and individual goals.

ACSM's Guidelines for Exercise Testing and Prescription, Tenth Edition,¹ is a rich source of updated and compiled information concerning the health benefits of exercise. Based on combined efforts by the American College of Sports Medicine and the American Heart Association, the following recommendations have been in place since 2007:⁷

- All healthy adults aged 18 to 65 years should participate in moderate-intensity aerobic physical activity (PA) for a minimum of 30 min on 5 days per week or vigorous-intensity aerobic activity for a minimum of 20 min on 3 days per week.
- Combinations of moderate- and vigorous- intensity activity can be performed to meet this recommendation.
- Moderate-intensity aerobic activity can be accumulated to total the 30 min minimum by performing bouts each lasting ≥ 10 min.
- Every adult should perform activities that maintain or increase muscular strength and endurance for a minimum of 2 days per week.
- Because of the dose–response relationship between PA and health, individuals who wish to further improve their fitness, reduce risk for chronic diseases and disabilities, and prevent unhealthy weight gain may benefit by exceeding the minimum recommended amounts of exercise.

RESEARCH PERSPECTIVE 20.2

Revised Physical Activity Guidelines for Americans

In November 2018, the U.S. Department of Health and Human Services released a revision to the *Physical Activity Guidelines for Americans*. The original guidelines, published in 2008, highlighted the fact that being physically active is one of the most important actions that people of all ages can take to improve their health. Although the primary audiences for the guidelines are policy makers and health professionals, individuals can also benefit from the information. The revised edition, based on science from the intervening 10 years, reconfirmed the importance of daily physical activity to reduce the risk of many chronic diseases and reaffirmed the goal of 150 minutes of moderate-intensity aerobic activity per week. However, recognizing that many adults may not be able to achieve this goal, the guidelines now also state that any

amount of daily activity—no matter how big or small—is beneficial to health. In that regard, the new guidelines eliminate the caveat that, in order to be beneficial, physical activity has to in bouts of at least 10 minutes. Other additions to the new guidelines include the following:

1. Information about the benefits of physical activity for older adults and people whose chronic conditions impede regular exercise
 2. Guidance for children as young as 3 years of age
 3. Contrasts between sedentary behavior and physical activity, and the risks associated with the former
 4. New knowledge about the health benefits of physical activity for brain health, certain cancers, and fall-related injuries
 5. Information about the benefits of exercise related to general well-being, daily functioning, and sleep
 6. Science-based strategies to get the population as a whole more active
-

During the first decade of the 21st century, the ACSM in collaboration with the American Medical Association (AMA) launched a major initiative to encourage health care providers to counsel their patients on the importance of physical activity in promoting and maintaining health and preventing disease. With the recognition that many physicians and associated health care workers have not had extensive education and training in this area, a formal program was developed to educate and train them in the basics of exercise prescription. The website ExerciselsMedicine.org offers the “Health Care Providers’ Action Guide” to guide clinicians in including physical activity in their patients’ overall health care plan.

VIDEO 20.1 Presents Bob Sallis with a physician’s perspective on the prescription of exercise.



Knowing the benefits of regular exercise, how should people begin exercise programs to improve their general health and fitness? The first step is deciding to take action. The next step is getting medical clearance.

Health Screening

Is a medical evaluation really necessary before one starts an exercise program? Although comprehensive medical screening is useful and desirable before exercise is prescribed, not all people need one. Many cannot afford the cost of such an evaluation, and the medical system is not prepared to provide this service for the total population, even if money were available. Also, medical evaluation before prescribing exercise for a population presumed healthy has not been proven to reduce the medical risks associated with exercise.

Most sports medicine organizations, including the ACSM, agree that a preparticipation health screening is encouraged for individuals who want to start or intensify an exercise program. However, medical evaluation is perceived as a significant barrier to starting an exercise program for many people. New guidelines clarify the level of screening suggested in the form of an algorithm, shown in [table 20.1](#).

TABLE 20.1A ACSM Preparticipation Screening Algorithm for Subjects Who Do Not Exercise Regularly

Health status	Medical clearance		Follow exercise prescription guidelines on progression*		
	Recommended	Not necessary	Light-intensity exercise (30%-<40% HRR)	Moderate-intensity exercise (40%-<60% HRR)	Vigorous-intensity exercise (≥60% HRR)
No CV, metabolic, or renal disease and no signs or symptoms of these		x	Recommended	Recommended	May progress to this intensity*
Known CV, metabolic, or renal disease and asymptomatic	x		Recommended, after medical clearance	Recommended, after medical clearance	May progress to this intensity*
Any sign or symptom of CV, metabolic, or renal disease	x		Recommended, after medical clearance	Recommended, after medical clearance	May progress to this intensity*

CV = Cardiac, peripheral vascular, or cerebrovascular disease. Metabolic disease = type 1 and 2 diabetes mellitus. Regular exercise is defined as performing planned, structured physical activity for ≥30 min at a moderate or higher intensity for at least 3 days per wk over the past 3 mo or longer. *See chapter 11 for details on exercise prescription.

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TABLE 20.1B ACSM Preparticipation Screening Algorithm for Subjects Who Exercise Regularly

Health status	Medical clearance		Follow exercise prescription guidelines on progression*		
	Recommended	Not necessary	Light-intensity exercise (30%-<40% HRR)	Moderate-intensity exercise (40%-<60% HRR)	Vigorous-intensity exercise (≥60% HRR)
No CV, metabolic, or renal disease and no signs or symptoms of these		x		May continue at this intensity	May continue or progress to this intensity*
Known CV, metabolic, or renal disease and asymptomatic		x		May continue at this intensity	
	x			May continue at this intensity	May continue this intensity after medical clearance
Any sign or symptom of CV, metabolic, or renal disease	x Stop exercising		May resume after medical clearance	May resume after medical clearance	May resume after medical clearance

CV = Cardiac, peripheral vascular, or cerebrovascular disease. Metabolic disease = type 1 and 2 diabetes mellitus. Regular exercise is defined as performing planned, structured physical activity for ≥30 min at a moderate or higher intensity for at least 3 days per wk over the past 3 mo or longer. *See chapter 11 for details on exercise prescription.

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RESEARCH PERSPECTIVE 20.3

Exercise and the Brain

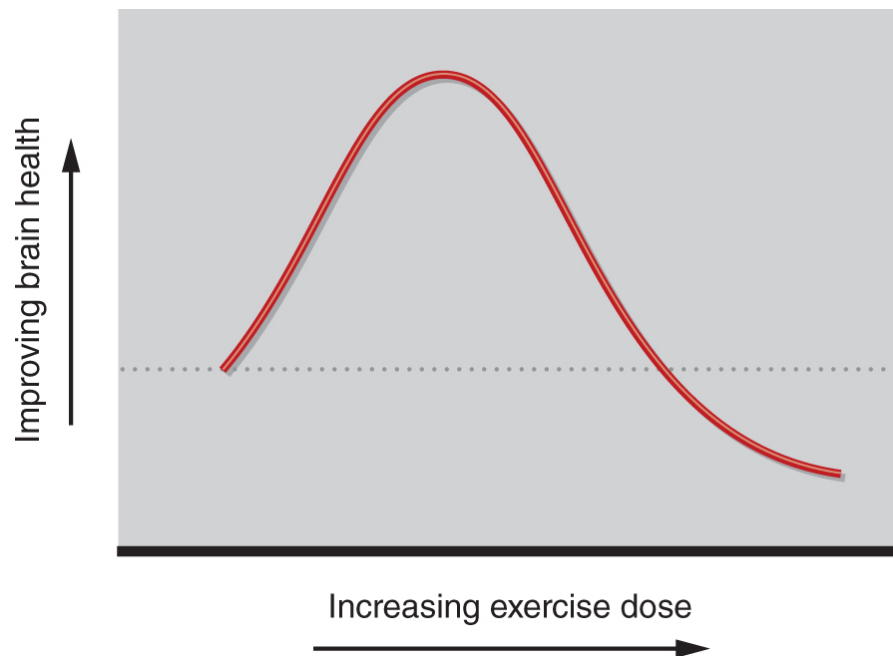
Over the past two centuries, life expectancy has nearly doubled, leading to an increasingly older population. Aging is the primary risk factor for chronic disease, including dementia and Alzheimer's disease, whose prevalence continues to steadily (and alarmingly) increase. Epidemiological studies suggest that the development of Alzheimer's disease may be related to modifiable risk factors, including physical activity and cardiovascular risk factors, in ~30% of patients. There is now convincing evidence linking habitual aerobic exercise to reductions in age-related cognitive decline. The precise mechanisms mediating exercise-induced improvements in brain structure and function are not clear. However, it appears likely that the cardiovascular adaptations that occur as a consequence of chronic physical activity may maintain brain health, reducing the risk of cognitive impairment and the development of dementia.

The cognitive benefits of exercise, including higher cognitive performance in memory and executive function, are greatest when regular exercise is started earlier in life and when aerobic exercise and strength training are combined.¹⁷ Habitual aerobic exercise training also causes structural adaptations in the brain, reducing the deteriorations in gray and white matter functioning that typically occur with aging. In addition, exercise preserves brain volume, particularly in brain areas associated with cognitive impairment (e.g., parietal lobe and visual cortex).

In addition, the beneficial effects that habitual physical activity has on cardiovascular function also likely improve brain health. The brain lacks intracellular energy storage and thus relies entirely on the vascular supply of oxygen and nutrients; therefore, exercise-induced improvements in cardiovascular control of cerebral blood flow play a critical role in maintaining normal brain structure and function. The brain is particularly sensitive to changes in blood pressure because of the low resistance of the cerebral vasculature. In this regard, regular aerobic exercise attenuates the stiffening of the central elastic arteries (the aorta and carotid arteries). Reductions in central arterial stiffness are accompanied by reductions in systolic blood pressure, an established risk factor for stroke and cognitive impairment, and blunt the transmission of excessive pulsatile blood pressure waves into the brain. Exercise also improves endothelial function in large conduit arteries, cerebral autoregulation and vasomotor reactivity, and substrate exchange and

waste clearance in the cerebral microcirculation, all of which ultimately improve brain health.

A 2014 epidemiological study also suggests that aerobic fitness reduces the risk of brain cancer mortality.¹⁸ In this study, the risk of brain cancer mortality was prospectively examined in the National Walkers' and Runners' Health Study cohort (~153,000 individuals, ~50 years old, followed for ~12 years). In this sample, 100 deaths were attributed to brain cancer as the primary cause. Within these individuals, the risk for brain cancer mortality was significantly reduced with greater amounts of regular exercise, with the most benefit incurred with more activity (e.g., running 12 to 25 km/week).



The potential hormetic relation between increases in exercise intensity and improvements in brain health. Aerobic exercise improves cognitive function and brain structure; however, strenuous exercise may have deleterious effects.

Interestingly, the relation between exercise intensity and brain health may not be linear; instead, this link appears hormetic (i.e., a biphasic dose-response relation) (see figure). That is, emerging evidence suggests that strenuous endurance exercise may actually *cause* brain injury if not enough recovery is allowed. The mechanisms mediating the adverse effects of exercise on the brain remain unclear but conceivably could be related to the increased risk of systemic catabolic burden, inflammatory responses, and risks of cardiovascular injury or event. However, future studies are necessary in order to more precisely define the optimal dose of exercise training that may prevent or slow the age-related functional and structural deteriorations in the brain. Nevertheless, aerobic exercise is clearly an effective strategy to improve cardiovascular health, thereby slowing cognitive decline and the

onset of dementia in older adults, improving the quality of life and extending the health span.

Health-Related Fitness Testing

Proper measurement of various aspects of physical fitness is common in preventive medicine, laboratory research, and rehabilitation and physical therapy programs.

While some of these tests are done for the purpose of diagnosing or documenting diseases such as coronary artery disease, the focus here is on the use of these tests to establish an individual's general health status and fitness level.

Body Composition

Increased adiposity is associated with a variety of diseases and disabilities. Central or abdominal adiposity has a particularly strong association. Height and weight measurements alone do not adequately capture this risk nor are they generally related to fitness level. Rather, determination of the components of the two-compartment model—fat mass and fat-free mass—can provide valuable information about health. Changes in these two components with exercise and diet are hallmarks of their effectiveness and health benefits. Techniques used for measurement of body composition are covered in detail in [chapter 15](#).

Muscular Strength and Endurance

The definitions and principles of muscular strength and endurance are covered in [chapter 9](#). These fitness components can be assessed in a number of valid ways in the laboratory or clinic. Muscular strength is defined as the maximal force that can be produced by a muscle or muscle group. Traditionally, the 1RM test has been the standard for measuring movement-specific strength, but 5RM and 10RM tests have also been proposed.¹ Grip strength is an important construct for activities of daily living and a good predictor of functional status and premature mortality in older individuals.

Muscular endurance, on the other hand, refers to the ability of a muscle or muscle group to either (1) sustain repeated contractions over a period of time or (2) maintain a given percentage of 1RM for a period of time. Perhaps the most common field test for muscular

endurance is the push-up test, i.e., the maximal number of properly performed push-ups that can be done in a fixed period of time (usually 1 or 2 min). Sit-up or curl-up tests are not recommended because they only moderately relate to abdominal endurance and may cause lower back injury.¹

Flexibility

The ability to move a joint through a complete range of motion is called flexibility, and tests of flexibility usually measure degrees of movement around that joint. While the flexibility of many joints can be measured, the most common flexibility test is the sit-and-reach test, or trunk flexion test. This test measures hamstring and lower back flexibility. Flexibility tests are unique in that women almost routinely score better than men of a similar age.

Cardiorespiratory Fitness

As mentioned throughout this book, the standard measure of cardiorespiratory fitness, or aerobic capacity, is maximal oxygen consumption, or $\dot{V}O_{2\max}$. $\dot{V}O_{2\max}$ can be estimated from field tests (such as the time it takes to complete a 12 min walk or run) or from a submaximal or maximal **graded exercise test (GXT)**. GXTs can be performed on a treadmill or a cycle ergometer or using a step test.

Submaximal tests typically involve measuring HR at several exercise intensities and extrapolating to $HR_{\max}$. This process is illustrated in [figure 8.2](#). Maximal GXTs push subjects to the point of volitional fatigue and often directly measure oxygen uptake at this maximal work load. GXTs are also used as clinical tools for the diagnosis or verification of coronary artery disease, as discussed next.

Clinical Graded Exercise Testing

Exercise testing alone can be used for determining readiness for exercise and activity or assessing the effectiveness of therapy as well as for other clinical decision-making situations, but it is rarely used in isolation for diagnostic purposes. When used for diagnostic purposes, a clinical GXT is usually used in conjunction with other procedures, such as echocardiography, pharmacological challenges using specific cardiovascular drugs or agents, or radionuclide scans. Diagnostic

testing may also include echocardiography and radionuclide imaging to obtain anatomical and physiological images of the myocardium. These techniques improve the specificity and the sensitivity of the test. A 12-lead **exercise electrocardiogram (ECG)** (figures 20.1 and 20.2) usually accompanies the GXT. Exercise blood pressures and other measures of physiological and even psychological responses (rating of perceived exertion and pain scales, for example) may also be obtained during these assessment procedures.

The ECG and blood pressure are monitored as the person progresses from low-intensity exercise such as slow walking to more intense exercise and sometimes, but rarely, up to maximal-intensity exercise. Maximal-intensity exercise is, again, rarely used or seen with current methods of diagnostic testing. Exercise testing is, however, generally progressive; that is, the rate of work is progressively increased every 1 to 3 min until some usually predetermined heart rate (commonly 85% of estimated maximum) is achieved. The ECG is monitored to detect heart rhythm and electrical conductivity abnormalities. Blood pressure is monitored to determine whether there is a normal increase in systolic blood pressure and little or no change in diastolic blood pressure as the rate of work progresses from low intensity to maximal or near-maximal levels. It is also important for the clinician to interact with the subject, observing signs and symptoms during and immediately following the exercise test, such as chest pain or pressure (angina), unusual shortness of breath, light-headedness or dizziness, and inappropriate heart rate response.

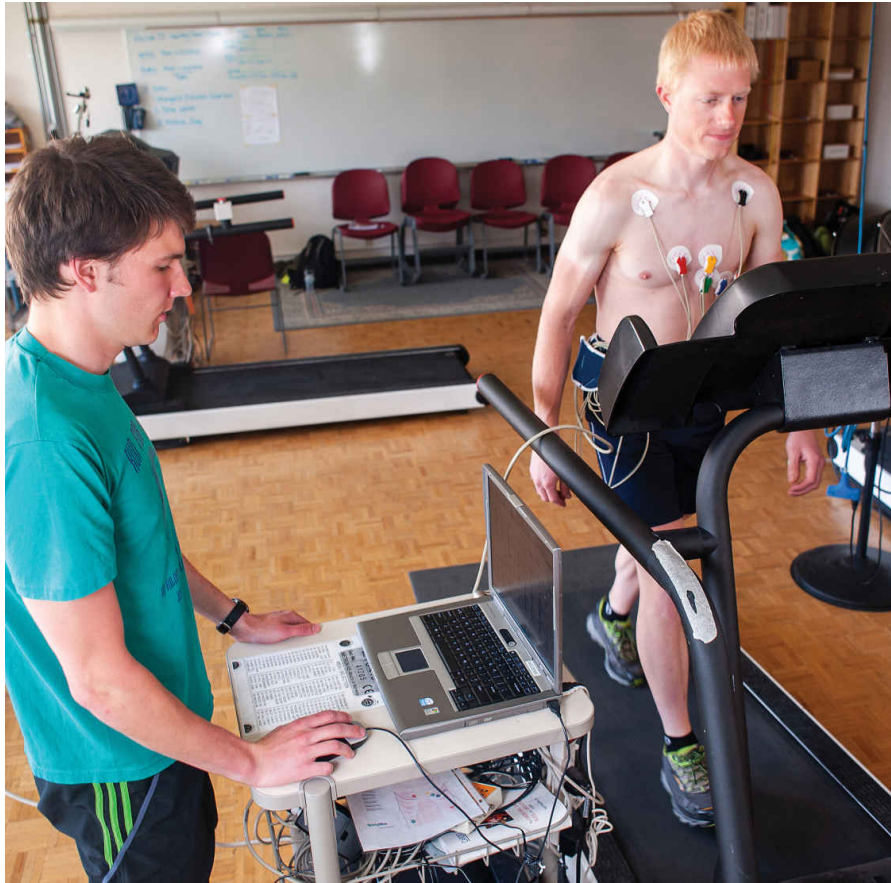


FIGURE 20.1 Obtaining an exercise electrocardiogram during a clinical graded exercise test.

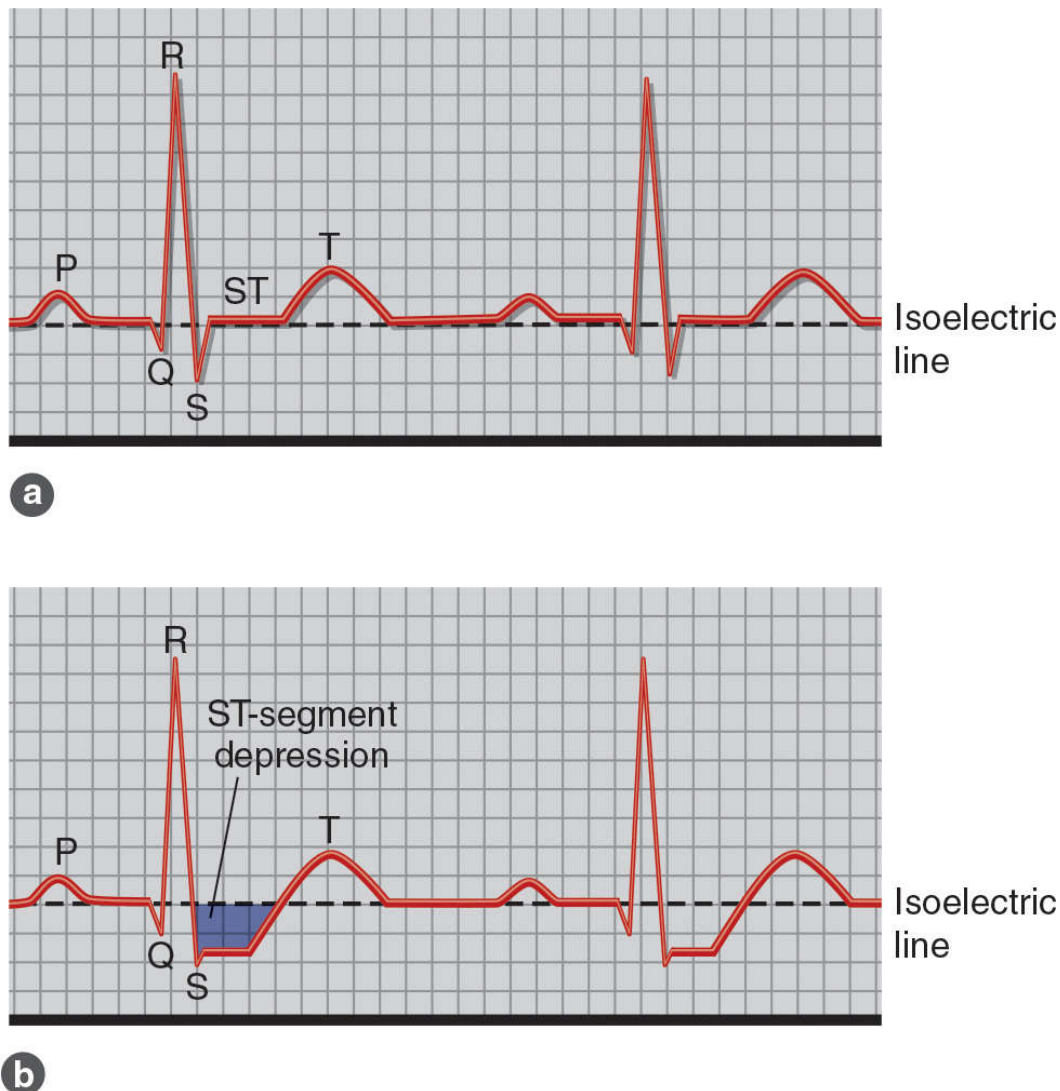


FIGURE 20.2 Illustration of (a) a normal electrocardiogram (ECG) and (b) an ECG with ST-segment depression, suggestive of the presence of coronary artery disease.

Results from a diagnostic GXT, the ECG, the echocardiogram, the images, and the physiological measurements are all considered in assessing the test as either positive, negative, or equivocal. Positive or equivocal tests usually lead to subsequent diagnostic procedures such as coronary angiograms, other cardiovascular testing procedures, or medical management to determine the etiology and origin of the signs and symptoms that brought the person to the physician in the first place. Typically, a person with abnormal changes in a diagnostic stress test is referred for more direct clinical procedures.

To determine the accuracy of exercise ECG results, the sensitivity, specificity, and predictive value of an exercise test must be considered. **Sensitivity** refers to the exercise test's ability to correctly identify people who have the disease in question, such as CAD. **Specificity** refers to the test's ability to correctly identify people who do not have the disease. And the **predictive value of an abnormal exercise test** refers to the accuracy with which abnormal test results reflect presence of the disease.

Unfortunately, both the sensitivity and the predictive value of an abnormal exercise test for detecting CAD are relatively low in healthy people who have no symptoms of this disease. Past studies reveal that sensitivity averages about 66%, indicating that 66% of those with CAD are correctly identified by exercise ECGs as having the disease. Conversely, 34% of those with the disease are incorrectly diagnosed as disease-free based on exercise ECGs. Using exercise testing on a population that has a high probability of having CAD raises the sensitivity. Thus, it is now relatively rare to see a GXT performed for purely diagnostic purposes because GXTs are of limited value in screening young, apparently healthy individuals for coronary artery disease.

VIDEO 20.2 Presents Bob Sallis on whether a physician should be consulted before beginning moderate exercise.



In Review

- Before beginning any exercise program, anyone who is considered to be at a high risk for CAD should be screened according to the ACSM screening algorithm.

- Health-related fitness tests include measures of body composition, muscular strength and endurance, flexibility, and cardiorespiratory fitness.
 - The most recent ACSM guidelines should be followed for each phase of the evaluation, and the physician should be consulted about the proposed exercise activity in case there are any medical contraindications.
 - Test sensitivity refers to the test's ability to correctly identify people with a given disease. Test specificity refers to its ability to correctly identify people who do not have the disease. The predictive value of an abnormal exercise test refers to the accuracy with which the test reflects presence of the disease in a given population.
 - The most recent ACSM guidelines state that a medical examination and exercise test might not be necessary if moderate exercise is undertaken gradually, without competition, in people without symptoms of cardiovascular, pulmonary, and metabolic disease.
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Exercise Prescription

The **exercise prescription** involves six basic factors:

1. Mode (type) of exercise
2. Frequency of participation
3. Intensity of the exercise bout
4. Duration (time) of each exercise bout
5. Volume of exercise
6. Progression

In the ACSM's newest guidelines,¹ the acronym FITT-VP is used to denote frequency, intensity, type, time, volume, and progression.

We outlined early in this chapter the recommended FITT-VP characteristics documented to promote health and minimize the risk of chronic disease. In this section, we assume that the goal of the exercise program is to improve cardiorespiratory endurance, including making small changes to sedentary behavior such as promoting breaks from sitting. Since the prescription of resistance training programs is discussed in detail in [chapter 9](#), we will only briefly mention the inclusion of resistance training as a part of a total exercise program later in this chapter. The focus of this section is on aerobic training. Also, the information in this section is not appropriate

for designing training programs for competitive endurance athletes, which is covered in [chapter 9](#).

A minimum threshold for frequency, duration, and intensity of exercise must be reached before aerobic benefits are obtained. But, as we have discussed elsewhere, individual responses to any given training program are highly variable, so the threshold required differs from one person to the next. If we use exercise intensity as an example, the ACSM defines the lower end of moderate training intensity as 40% of one's heart rate reserve (HRR) or $\dot{V}O_{2\max}$.¹ Although this recommendation is appropriate for most healthy adults, some low-fit individuals might improve their aerobic capacities at intensities below 40% of their $\dot{V}O_{2\max}$. Each individual's threshold for frequency, duration, and intensity must be exceeded to achieve gains in cardiorespiratory endurance, and this threshold is likely to increase as cardiorespiratory endurance improves.

Exercise Mode

The prescribed exercise program should focus on one or more **modes**, or types, of cardiorespiratory endurance activities. Traditionally, the activities prescribed most frequently are

- walking,
- jogging,
- running,
- hiking,
- cycling,
- rowing, and
- swimming.

Because these activities do not appeal to everyone, alternative activities have been identified that promote similar cardiorespiratory endurance gains. Spinning, cardiovascular (“cardio”) endurance group exercise, and combination resistance training–cardiorespiratory endurance training programs are excellent substitutes for the modes of cardiorespiratory endurance exercise listed. The benefits of HIIT training have been discussed throughout the book.

Individuals should select activities that they enjoy and are willing to continue throughout life. Exercise must be regarded as a lifetime

pursuit because, as we saw in [chapter 14](#), the benefits are soon lost if participation stops. Motivation is probably the most important factor in a successful exercise program. Selecting an activity that is fun, provides a challenge, and can produce needed benefits is one of the most crucial tasks in exercise prescription. Having several different activities to pursue is also wise in case of inclement weather, travel, or other barriers. Other considerations include geographic location, climate, and availability of equipment and facilities. Home exercise is useful when people are homebound either because of responsibilities, such as child rearing, or because of weather considerations such as heat, humidity, cold, rain, ice, and snow.

RESEARCH PERSPECTIVE 20.4

Golf Is (Probably) Good for Your Health

The potential of golf to provide an opportunity for physical activity, and the associated health benefits, is perhaps readily apparent. This may be particularly true for middle-aged and older adults, who are generally less active than younger adults. However, limited studies have comprehensively examined the relation between golf and health. Thus, a group of investigators recently undertook a scoping review to map the available evidence and identify the existing gaps in knowledge regarding the impact of golf on health.¹² Scoping reviews summarize a range of evidence in order to convey the breadth and depth of a field. In contrast to a systematic review, authors generally do not assess the quality of the included studies. Overall, this scoping review identified 301 eligible studies that assessed the effects of golf on physical and mental health. Although the scoping review cannot determine causative effects, the reports included provide consistent evidence for a positive association between golf and physical and mental health. Priority areas for future research include the effects of golf on mental well-being, the relation of golf to muscle strength and function, balance and fall prevention, and health behaviors among golfers.

Exercise Frequency

Research studies on exercise frequency show that 3 days per week of vigorous activity or 5 days per week of moderate activity are optimal. This does not mean that 6 or 7 days per week won't give additional benefits, but simply that for the health-related benefits, the optimal gain is achieved with a time investment of 3 to 5 days per

week. Obviously, additional days above the 3- or 4-day frequency are beneficial for weight loss, but this frequency should not be encouraged until the exercise habit is firmly established and the injury risk is reduced.

Exercise Intensity

The intensity of the exercise bout appears to be the most important factor for developing cardiorespiratory endurance. How hard must people push themselves to gain benefits? Former athletes immediately recall the exhaustive workouts they endured to condition themselves for their sport. Unfortunately, this concept also gets carried over into the exercise programs they pursue for health benefits. Evidence now suggests that a modest training effect can be seen in some people through training at intensities of 40% of their $\dot{V}O_{2\max}$. Generally, these intensities are most effective for people at lower levels of aerobic fitness. For most, however, the appropriate intensity appears to be between 40% and 90% of HRR. An upper level for intensity depends on the purposes and goals of the training program.

The ACSM defines a moderate training intensity of 40% to 59% of one's heart rate reserve (HRR) or $\dot{V}O_{2\max}$ and vigorous exercise as 60% to 90% of HRR or $\dot{V}O_{2\max}$.¹

Multiple studies have clearly demonstrated that high-intensity interval training (HIIT) of low volume and very high intensity can markedly increase cardiorespiratory endurance. HIIT training has metabolic effects similar to longer, lower-intensity training. Substantial increases in muscle oxidative capacity, cardiometabolic variables associated with the protective effects of exercise, and cardiorespiratory endurance performance have been obtained in a training period as short as 10 min per day for 2 weeks.⁶

Exercise Duration

Duration and intensity of exercise are closely linked. Several studies have demonstrated improvement in cardiovascular conditioning with endurance exercise periods of 20 to 30 min per day at an appropriate intensity. Again, "optimal" is used here to reflect the greatest return for time invested, and the specified time refers to the time during which one is at one's appropriate exercise intensity. Similar improvements in

cardiorespiratory endurance can be gained with either a short-duration, high-intensity program or a long-duration, low-intensity program if the minimal threshold for both duration and intensity is exceeded. Similar benefits are also gained whether the daily endurance training session is conducted in multiple shorter bouts (e.g., three 10 min bouts) or a single long one (e.g., one 30 min bout). Longer bouts are most often prescribed to facilitate weight loss or maintenance of weight loss.

Exercise Volume

It is appropriate to specify the weekly volume of exercise as MET-minutes (METs × total weekly minutes) or calories expended per week. MET stands for metabolic equivalent, a system for quantifying the intensity of activities that is discussed in more detail later in this chapter. Using MET-minutes per week as a single measure that combines intensity, frequency, and volume allows the exercise prescription to be progressed in a more deliberate, quantifiable manner (and more cautiously when appropriate). The ACSM recommends ≥500 to 1,000 MET-minutes/week or 150 min/week of moderate-intensity exercise. Caloric expenditures of 1,000 to 1,500 per week are close to those volume levels, depending on weight and the intensity of the exercise.

Progression

The rate at which an individual progresses through an exercise program is variable and must be individualized. Recall from [chapter 9](#) that overload is a general principle of exercise training. Progression may involve changing one, or several, of the individual FITT-VP components.

Breaks in Prolonged Sitting

Prolonged sitting is a fact of life in our everyday world. Watching TV, sitting at work in front of computer screens all day, driving and commuting, eating, even having coffee with friends all involve sitting. Sedentary activities make up almost 60% of the day for most Americans and, if sleeping is included, may exceed 75% of the day.

Important research on sedentary behavior (defined as physical activity <1.5 METs) has described a sedentary physiology that

promotes chronic disease. Pathophysiological conditions that are common to cardiovascular disease, obesity, diabetes, and other chronic diseases, in fact, are similar to those induced by being sedentary. As few as 30 min of continuous sitting increases insulin resistance, negatively affects lipid and carbohydrate metabolism, shifts ongoing metabolism more toward glucose, promotes storage of fat and changes in muscle fiber type, and, through those mechanisms, promotes a systemic, subacute inflammation.² All of these changes are consistent with and similar to those found in the chronic diseases mentioned.

Since it is clear that prolonged and continuous sitting causes these changes, the next question is how to moderate or prevent them. Does regular exercise do that, or do breaks in sitting do that? Here's what is known: Regular exercise, while clearly preventive with regard to chronic disease, does not change those effects of sedentary behavior. Breaks in sitting or sedentary behavior do, in fact, change that physiology. As little as 2 min of low-intensity (2.0-2.5 METs) activity can positively affect such physiological variables as insulin resistance and glucose metabolism.¹⁴ The evidence is compelling enough that people with sedentary lifestyles should consider simply getting up and moving around periodically to break up the duration of sitting or other sedentary behavior.

In Review

- Six basic elements of an exercise program are exercise mode, frequency, intensity, duration, volume, and progression. A minimum threshold for the frequency, intensity, and duration must be met to yield significant cardiorespiratory endurance benefits, and this threshold is quite variable from one individual to another.
- The program should include one or more cardiorespiratory endurance activities. If the activity involves competition, preconditioning with a standard endurance activity is recommended before sport participation begins in order to bring the person up to an appropriate level of fitness.
- Activities must be matched with individual needs and likes so that motivation can be maintained.
- Optimally, exercise should be performed daily or on most days of the week. Exercise should begin with three or four sessions per week and then progress to

more if desired.

- Exercise duration of 30 to 60 min working at moderate intensity or 20 to 60 min at vigorous intensity is recommended, but the key is reaching the threshold for combined duration and intensity, which can be expressed as MET-minutes.
 - For most people, intensity should be at least 40% to 59% HRR (moderate) but can range from 40% to 90%. However, health benefits might occur at intensities lower than those needed for cardiorespiratory endurance conditioning in deconditioned people and can also occur at very high intensities.
 - Research suggests that prolonged sitting has negative impacts on a person's health, even if the person does regular exercise. Frequent breaks from sitting are recommended as part of an exercise prescription for health.
-

Monitoring Exercise Intensity

Exercise intensity can be quantified on the basis of the training heart rate (THR), the metabolic equivalent (MET), or the rating of perceived exertion (RPE). Let's examine each of these and their strengths and weaknesses in quantifying exercise intensity.

Training Heart Rate

The concept of **training heart rate (THR)** is based on the linear relation between heart rate and $\dot{V}O_2$ with increasing rates of work, as shown in [figure 20.3](#). When people are exercise tested, their heart rate and $\dot{V}O_2$ values are obtained each minute and plotted against each other. The THR is established through use of the heart rate that is equivalent to a set percentage of the $\dot{V}O_{2max}$. For example, if a training level of 75% $\dot{V}O_{2max}$ is desired, 75% of $\dot{V}O_{2max}$ is calculated ($\dot{V}O_{2max} \times 0.75$), and the heart rate corresponding to this $\dot{V}O_2$ then is selected as the THR. An important point is that the exercise intensity necessary to achieve a given percentage of $\dot{V}O_{2max}$ does not necessarily result in a matching percentage of maximal heart rate (HR_{max}). As an example, a THR that is set at 75% of the $\dot{V}O_{2max}$ represents an intensity of approximately 65% to 90% of the HR_{max} (see [figure 20.3](#)).

One can also establish the THR by using the heart rate reserve (HRR), also known as the Karvonen method. **Heart rate reserve** is

defined as the difference between $HR_{\max}$ and the resting heart rate (HR_{rest}):

$$\text{Maximal HRR} = HR_{\max} - HR_{\text{rest}}$$

With this method, we calculate the THR by taking a given percentage of the maximal HRR and adding it to the resting heart rate. Let's consider an example. For 75% of maximal HRR, the equation would be as follows:

$$THR_{75\%} = HR_{\text{rest}} + 0.75 (HR_{\max} - HR_{\text{rest}}).$$

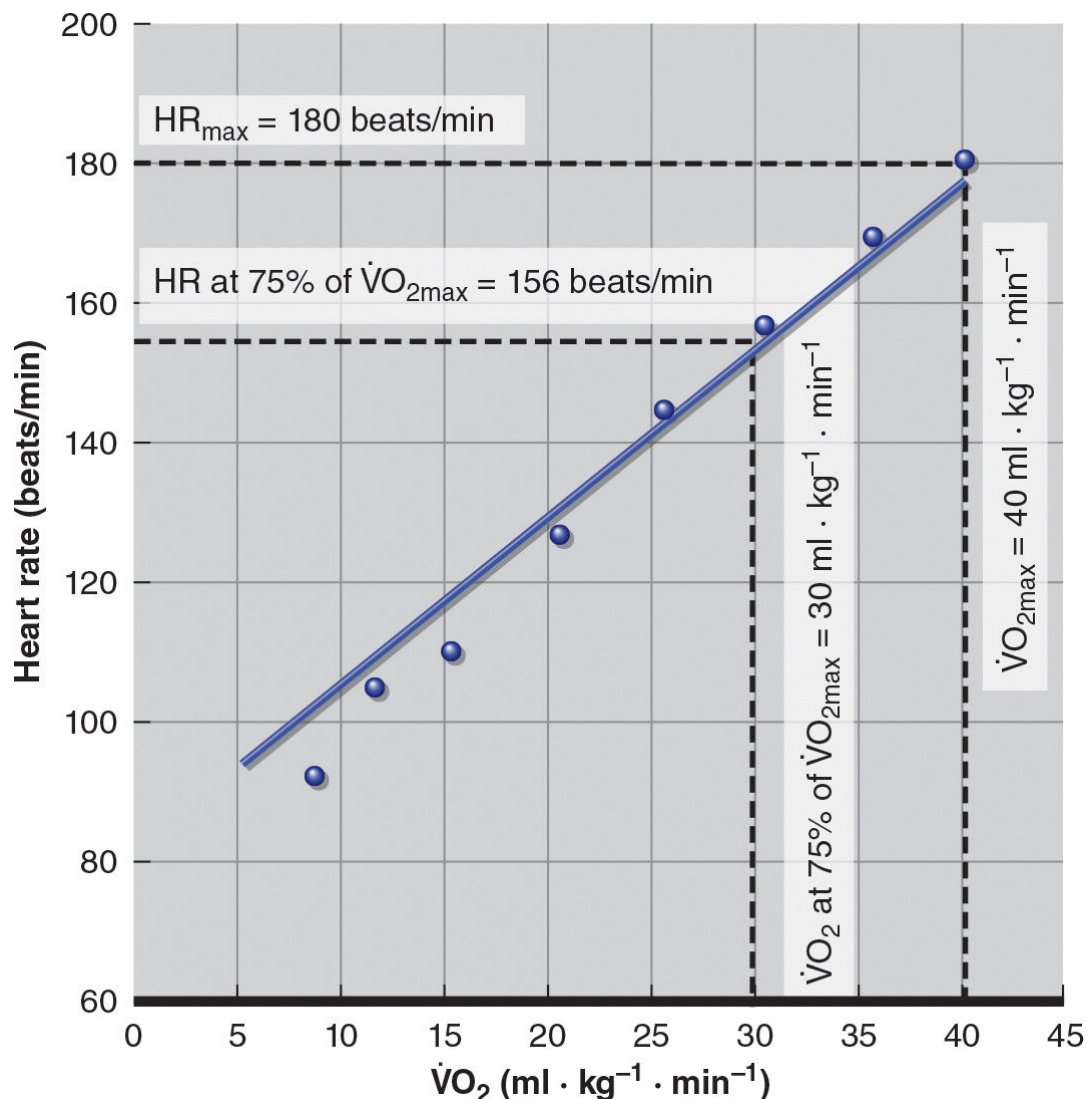


FIGURE 20.3 The linear relation between heart rate and oxygen consumption ($\dot{V}O_2$) with increasing rates of work and the heart rate equivalent to a set percentage (75%) of $\dot{V}O_{2\max}$.



Heart rate reserve adjusts the THR so that the THR as a specific percentage of the maximal HRR is nearly identical to the heart rate equivalent of that same percentage of $\dot{V}O_{2\max}$ at moderate to high intensities. Thus, a THR computed as 75% of maximal HRR is approximately the same as the heart rate corresponding to 75% of the $\dot{V}O_{2\max}$. However, there may be a substantial difference between the two at low intensities.¹⁶

Training Heart Rate Range

An appropriate exercise intensity can be established by setting a heart rate range rather than a single THR value. This is a more sensible approach because exercising at a set percentage of $\dot{V}O_{2\max}$ can place people above their lactate threshold, making it difficult for them to train for an extended period of time. With the THR range concept, low and high values are established that can ensure a training response. Individuals may choose to exercise at the low end of the HRR range or to progress through the range as comfort dictates. To illustrate this, using the HRR method for establishing the THR, consider the following example. A 40-year-old man has a resting heart rate of 75 beats/min and a maximum heart rate of 180 beats/min, and he is advised to exercise within a THR range of 50% to 75% HRR. His THR range would be as follows:

$$\begin{aligned}\text{THR}_{50\%} &= 75 + 0.50 (180 - 75) \\ &= 75 + 53 = 128 \text{ beats/min}\end{aligned}$$

$$\begin{aligned}\text{THR}_{75\%} &= 75 + 0.75 (180 - 75) \\ &= 75 + 79 = 154 \text{ beats/min}\end{aligned}$$

This same THR range method can be used if one estimates $\text{HR}_{\max}$ $[208 - (0.7 \times \text{age})]$ without losing much accuracy if true $\text{HR}_{\max}$ has not been determined.

Heart rate is the preferred method for monitoring exercise intensity because it is highly correlated to the work of the heart (or stress on the heart) and allows for a progressive increase in the rate of training with improvements in fitness to maintain the same THR. The use of the HRR (Karvonen) method is the preferred method for establishing the THR range. When one is prescribing exercise intensity, it is appropriate to establish a THR range, with exercise starting at the low end of the range and progressing to the upper end of the range over time.

The $\dot{V}O_{2\max}$ Reserve Method

The ACSM also advocates a slightly different approach to prescribing exercise intensity. Exercise intensity is prescribed based on what has been termed the $\dot{V}O_2$ reserve method ($\dot{V}O_2R$). Instead of prescribing exercise at a given percentage of $\dot{V}O_{2\max}$, one bases the prescription on a given percentage of the $\dot{V}O_2R$, where $\dot{V}O_2R$ is defined as $\dot{V}O_{2\max} - \dot{V}O_{2\text{rest}}$. This also can be thought of as the $\dot{V}O_{2\max}$ reserve. As an example, with a $\dot{V}O_{2\max}$ of $40 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ and a $\dot{V}O_{2\text{rest}}$ of $3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$,

$$\begin{aligned}\dot{V}O_2R &= 40 - 3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \\ &= 36.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}.\end{aligned}$$

To prescribe an exercise intensity range between 60% and 75% of $\dot{V}O_2R$, we simply multiply $\dot{V}O_2R$ by 60% and 75%:

$$\begin{aligned}\dot{V}O_2R_{60\%} &= 36.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \times 0.60 \\ &= 21.9 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\end{aligned}$$

$$\begin{aligned}\dot{V}O_2R_{75\%} &= 36.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \times 0.75 \\ &= 27.4 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\end{aligned}$$

The major advantage of using the $\dot{V}O_2R$ technique is that one now has an equivalency between the percentage of the maximal HRR and the percentage of $\dot{V}O_{2\max}$ reserve. There is a potential problem with

this technique, however, in that using $3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ as a standard value for $\dot{V}\text{O}_{2\text{rest}}$ assumes that everyone has the same resting value. This, in fact, is not the case. Further, in one study, a large sample of women ($n = 642$) and men ($n = 127$) were found to have average $\dot{V}\text{O}_{2\text{rest}}$ values of 2.5 and 2.7 $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, respectively. The range of values varied from 1.6 to 4.1 $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$.⁵ Ultimately, however, the $\dot{V}\text{O}_2\text{R}$ technique mirrors the HRR method for prescribing exercise because the two values are so closely related.

Metabolic Equivalents

Exercise intensity also has been prescribed on the basis of the **metabolic equivalent (MET)** system. The amount of oxygen the body consumes is directly proportional to the energy expended during physical activity. In this system, it is assumed that the body uses approximately 3.5 ml of oxygen per kilogram (2.2 lb) of body weight per minute ($3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) at rest. As noted earlier, resting oxygen consumption varies among individuals, making this assumption not entirely accurate. The MET system, however, is based on this value, and the resting metabolic rate value of $3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ is referred to as 1 MET. All activities can be classified by intensity according to their oxygen requirements. An activity that is rated as a 2-MET activity would require two times the resting metabolic rate, or $7 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, and an activity that is rated at 4 METs would require approximately $14 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. Some activities and their MET values are presented in [table 20.2](#).

These values are only approximations because of the potential error in using a standard value of $3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ as a constant resting value. Further, metabolic efficiency varies considerably from one person to the next, and even in the same individual. Although the MET system is useful as a guideline for training, it fails to account for changes in environmental conditions, and it does not allow for changes in physical conditioning as discussed in the previous section. The MET system for exercise prescription is, however, very useful for determining someone's potential ability to perform physical activity that may not be exercise. For example, a deconditioned person who has just started an exercise training program and is

walking on a treadmill at 3.5 to 4.0 mph (4.3-5.0 METs) might ask whether it would be appropriate to incorporate tennis doubles into the program. Using the MET chart, you can see that doubles tennis is in the range of 4.5 to 6.0 METs. Thus you may advise the person that it is within his or her aerobic capacity, but that some of the MET levels for doubles tennis may be above that aerobic capacity. Additionally, activities that require some level of skill may also be more locally fatiguing. Though these facts may not preclude the activity, they may make it more fatiguing than it would be if the person were more fit.

TABLE 20.2 Selected Activities and Their Respective MET Values

Activity	MET value	Activity	MET value
Rest and self-care activities			
Rest (supine)	1.0	Showering	2.0
Sitting	1.5	General grooming, standing	2.0
Eating	1.5	Dressing or undressing, standing	2.5
Bathing	1.5		
Home activities			
Knitting or hand sewing, light effort	1.3	Vacuuming (general, moderate effort)	3.3
Washing dishes	1.8	Making beds, changing linens	3.3
Ironing	1.8	Cleaning (scrubbing floor, washing car, washing windows)	3.5
Laundry, folding or hanging clothes	2.0-2.3	Sweeping, moderate effort	3.8
Cooking or food preparation	2.0-3.5	Moving furniture, carrying boxes	5.8
Machine sewing	2.8	Scrubbing floors on hands and knees, vigorous effort	6.5
Occupational			
Sitting tasks, office work, working at a computer	1.5	Construction (outside)	4.0
Driving a delivery truck, taxi, school bus, and so on	2.0	Hotel housekeeper	4.0
Cook, chef	2.5	Yard work	4.0
Standing tasks, light to moderate effort	3.0-4.5	Manual or unskilled labor	2.8-6.5
Custodial work	2.5-4.0	Farming, light to vigorous effort	2.0-7.8
Carpentry (general, light to moderate effort)	2.5-4.3	Firefighter on the job	6.8-9.0
Physical conditioning			
Walking			
2.5 mph, level	3.0	4.5 mph, level	7.0
3.5 mph, level	4.3	5.0 mph, level	8.3
4.0 mph, level	5.0	5.0 mph, 3% grade	9.8
Jogging or running on level surface			
4.0 mph	6.0	10.0 mph	14.5
6.0 mph	9.8	12.0 mph	19.0
8.0 mph	11.8	14.0 mph	23.0
Swimming			
Freestyle, vigorous effort	9.8	Breaststroke, recreational or training and competition	5.3-10.3
Freestyle, slow to moderate	5.8	Sidestroke, general	7.0
Backstroke, recreational or training and competition	4.8/9.5		
Cycling			
Leisure, 5.5 mph	3.5	Leisure, 14.0-15.9 mph (vigorous effort)	10.0
Leisure, 10.0-11.9 mph (slow, light effort)	6.8	Racing, 16.0-19.0 mph (vigorous effort)	12.0
Leisure, 12.0-13.9 mph (moderate effort)	8.0	Racing, >20 mph (vigorous effort)	15.8
Recreational activities			
Aerobic dance	5.0-7.3	General resistance training	3.5-6.0
Video game activities	2.3-6.0	Rowing machines	4.8-12.0
Stationary cycle ergometer	3.5-14.0	Water aerobics	5.3

Activity	MET value	Activity	MET value
Circuit training	4.3-8.0	Video exercise workouts, light to vigorous	2.3-6.0
Sport activities			
Archery	4.3	Rock or mountain climbing	5.0-8.0
Badminton	5.5-7.0	Roller skating	7.0
Basketball	6.0-9.3	Rugby	6.3-8.3
Bowling, lawn bowling	3.0-3.8	Skateboarding	5.0-6.0
Football, flag or touch	4.0-8.0	Soccer	7.0-10.0
Golf	4.8	Softball	5.0-6.0
Handball	12.0	Squash	7.3-12.0
Hockey, field	7.8	Table tennis (Ping-Pong)	4.0
Hockey, ice	8.0-10.0	Tennis, singles	7.3-8.0
Horseback riding	5.8-7.3	Tennis, doubles	4.5-6.0
Lacrosse	8.0	Volleyball	3.0-4.0
Orienteering	9.0	Volleyball, competitive	8.0
Racquetball	7.0-10.0	Volleyball, competitive beach	6.0

Data from B.E. Ainsworth et al., "2011 Compendium of Physical Activities: A Second Update of Codes and MET Values," *Medicine and Science in Sports and Exercise* 43, no. 8 (2011):1575-1581.

Ratings of Perceived Exertion

Ratings of perceived exertion (RPE) also have been proposed for use in prescribing exercise intensity. With this method, individuals subjectively rate how hard they feel they are working. A given numerical rating corresponds to the perceived relative intensity of exercise. When the RPE scale is used correctly, this system for monitoring exercise intensity has proven very accurate. On the **Borg RPE scale**, which is a rating scale ranging from 6 to 20, moderate exercise corresponds to an RPE of 12 to 14 (somewhat hard) and vigorous exercise to an RPE of 14 to 16 (hard).⁴ Initially, this sounds too simple. However, most people can use the RPE technique very accurately. Studies have shown that when people are asked to select a pace on a treadmill or a resistance on a cycle ergometer at a moderate or heavy intensity of exercise (see [table 20.3](#)), they are able to select a pace or resistance that gets their heart rates into the appropriate range. This is a more natural way to prescribe exercise and very efficient if the person is able to relate perceptions of intensity accurately.

One simple way of monitoring exercise intensity is referred to as the talk test, which has been used as an informal guideline for years. Scientists have now confirmed that the highest exercise intensity that just barely allows a person to talk comfortably while exercising is a very consistent method that correlates well with ventilatory threshold (see [chapter 8](#)) and is well within the THR range.¹³

[Table 20.3](#) compares the various methods for rating exercise intensity. Let's use them to determine a moderate exercise intensity.

As the table shows, one would want to work within a range of 40% to 60% if one is monitoring intensity using the HRR method. Using rating of perceived exertion, shown in the fourth column, this is equivalent to an RPE value of 12 to 13. All these values reflect moderate-intensity exercise. Perhaps the best use of RPE is to combine it with THR, allowing the exerciser to self-select exercise intensity by both heart rate and comfort level. Thus, self-selecting the comfort level is also part of the exerciser's repertoire. The relationship between HRR and RPE can be so close (assuming the exerciser uses the RPE scale honestly and correctly) that most people can control most of their exercise by RPE and not have to take heart rate.

TABLE 20.3 Classification of Exercise Intensity Based on 20 to 60 min of Endurance Activity: Comparing Three Methods

Classification of intensity	Relative intensity		Rating of perceived exertion
	HR _{max}	HRR, $\dot{V}O_2R$	
Very light	<57%	<30%	<9
Light	57%-64%	30%-40%	9-11
Moderate	64%-76%	40%-60%	12-13
Vigorous	76%-96%	60%-90%	14-17
Near maximal to maximal	≥96%	≥90%	≥18

Data from American College of Sports Medicine (2018).

In Review

- Exercise intensity can be monitored on the basis of THR, metabolic equivalent, or RPE.
- Training heart rate can be established through use of the heart rate equivalent to a certain percentage of $\dot{V}O_{2max}$. It can also be determined using the HRR method, which takes a given percentage of maximal HRR and adds it to resting heart rate. With this method, the percentage of maximal HRR used corresponds to approximately the same percentage of $\dot{V}O_{2max}$ when a person is exercising at moderate to high intensities.
- A sensible approach is to establish a THR range to work within, rather than a single THR, attempting to establish the low end at an intensity below lactate threshold.
- The amount of oxygen consumed reflects the amount of energy expended during an activity. $\dot{V}O_2$ at rest has been assigned a value of $3.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, which

equals 1 MET. Activity intensities can be classified by their oxygen requirements as multiples of the resting metabolic rate.

- The RPE method requires that a person subjectively rate how difficult the work is, using a numerical scale that is related to exercise intensity. The subject looks at the standard scale to determine the appropriate number.
-

Exercise Programming

Once the exercise prescription has been determined, it is integrated into a total exercise program, which is generally only part of an overall health improvement plan. Individual exercise capacity varies widely even among people of similar ages and physical builds. For this reason, each program must be individualized based on results of physiological and medical tests and, if possible, individual needs and interests.

The total exercise program consists of the following activities:

- Warm-up and stretching activities
- Endurance training
- Cool-down and stretching activities
- Flexibility training
- Resistance training
- Neuromotor exercise
- Recreational activities

Generally, the first three activities are performed three or four times each week. Flexibility training can be included as part of the warm-up, cool-down, or stretching exercises, or it can be done at a separate time during the week. Resistance training is usually done on alternate days when endurance training is not performed; however, the two can be combined into the same workout.

Warm-Up and Stretching Activities

The exercise session should begin with a low-intensity, dynamic warm-up that includes some of the cardiorespiratory endurance activities and perhaps functional (neuromotor) activities that are part of the exercise program itself but at a much reduced intensity. Stretching (flexibility) exercises may be included in the warm-up. Such a warm-up period gradually increases both heart rate and

breathing, and prepares the exerciser for efficient and safe functioning of the heart, blood vessels, lungs, and muscles during the more vigorous exercise that follows. For example, someone who trains by running might start with 5 to 10 min of a light jog, some lunges or light-resistance band work, and then stretch before beginning the training run.

Endurance Training

Physical activities that develop cardiorespiratory endurance are the heart of the exercise program. They improve both the capacity and efficiency of the cardiovascular, respiratory, and metabolic systems. These activities may also help one control or reduce body weight. Activities such as walking, jogging, running, cycling, swimming, rowing, group cardiorespiratory endurance and other exercise activities, and hiking are good endurance activities. Sports such as handball, racquetball, tennis, badminton, and basketball also have aerobic potential if they are pursued vigorously. Activities such as golf, bowling, and softball are generally of little value for developing cardiorespiratory endurance, but they are fun, have definite recreational value, and may offer health-related benefits as physical activity. For these reasons, such activities certainly have a place in the overall exercise program.

Cool-Down and Stretching Activities

Every endurance exercise session should conclude with a cool-down period. The best way to accomplish cool-down is to slowly reduce the intensity of the endurance activity during the last several minutes of a workout. After running, for example, a slow, restful walk for several minutes helps prevent blood from pooling in the extremities. Stopping abruptly after an endurance exercise bout causes blood to pool in the legs and can result in dizziness or fainting. Also, catecholamine levels may be elevated during the immediate recovery period, which can lead to a heart dysrhythmia.

After the cool-down period, stretching exercises can be performed to facilitate increased flexibility.

Flexibility Training

Flexibility exercises usually supplement exercises performed during the warm-up or cool-down period and are useful for those who have poor flexibility or muscle and joint problems, such as lower back pain that may result from musculoskeletal imbalances. These exercises should be performed slowly. Quick stretching movements are potentially dangerous and can lead to muscle pulls or spasms. At one time it was recommended that these exercises be performed before the endurance conditioning period. However, recently it has been hypothesized that the muscles, tendons, ligaments, and joints are more adaptable and responsive to flexibility exercises when they are done after the endurance conditioning phase. Research has yet to confirm this hypothesis.

VIDEO 20.3 Presents Malachy McHugh on the timing and duration of stretching, and the role of stretching in injury prevention.



Resistance Training

The importance of resistance training as part of a general health and fitness exercise program has been clearly established. Many health-related benefits can be obtained from resistance training. The ACSM has included resistance training in its recommendations for a general health and fitness program.¹

Recall from [chapter 9](#) that the maximum amount of weight one can lift successfully only one time is the 1-repetition maximum, or 1RM. When people begin a resistance training program, they should start with a weight that is between 40% and 60% of their maximal strength, or 1RM, for each lift. The level will depend on age, conditioning level

(especially previous resistance training), and goals. Generally, again depending on goals and experience, between 8 and 12 repetitions are recommended for improvement of strength and power in most adults. Up to 15 to 20 repetitions of a specific weight in two to four sets may be recommended if goals or conditioning levels are significantly different from those common to most adults.

When a given weight brings the exerciser to fatigue, that is called “repetition max” (or rep max). People should try to achieve as many repetitions as possible during the second and third sets, but the number of repetitions they can complete in these last sets may decrease as their muscles become fatigued. As muscular strength and endurance increase, the number of repetitions they can complete per set will increase. When people reach 15 repetitions on the first set, they are usually ready to progress to the next higher weight. This training technique is referred to as progressive resistance exercise, as discussed in [chapter 9](#).

One can perform two or three sets of each lift per day, 2 or 3 days per week, for weight control purposes. Strength gains, however, appear to be fully achieved with just one set per day in previously untrained people.¹⁴ People should select a variety of exercises that tax most or all of the major muscle groups of the upper body, trunk, and lower body. If time is a factor, it is better to reduce the number of sets to one or two and maintain a full-body workout.

Neuromotor Exercise

Neuromotor exercise includes such aspects of fitness as balance, neuromotor engagement of musculature, proprioceptive training, gait, and perhaps even core training.¹ This is also called *functional training* by many people. Many physical activities can be included; most are multijoint exercises using body weight or light hand weights and other devices (e.g., kettle bells, bands, weighted balls) that are incorporated into movements such as lunges, lifts, and bending-twisting. These exercises are aimed at improving our ability to perform everyday activities that we all do in order to live. They also train us to engage muscle groups—for example, gluteal or abdominal muscles—that we tend to become less able to use as we become sedentary and stop using a wide variety of muscles.

There is little research on the effects, the benefits, or the exact amount of functional exercise that is beneficial, though some initial results indicate that it may help in fall prevention in older adults and reduce injury in athletes. However, many professionals prescribe functional exercise for healthy adults, and these exercises should be considered for inclusion in programs for healthy adults and older adults.

Recreational Activities

Recreational activities are important to any comprehensive exercise program. Although people engage in these activities primarily for enjoyment and relaxation, many recreational activities can also improve health and fitness. Activities such as hiking, tennis, handball, squash, and certain team sports fall into this category. Guidelines for selecting these activities include the following:

- Can you learn or perform the activities with at least a moderate degree of success?
- Do the activities include opportunities for social development, if that is desired?
- Are the costs associated with participation reasonable and within your budget?
- Are the activities varied enough to maintain your continued long-term interest?
- Given your current age and health status, is the given activity safe for you?

Many excellent opportunities exist for people who have no recreational hobbies or activities but would like to become involved in some way. Local public recreation centers, park districts, YMCAs, YWCAs, churches, some public schools, community colleges, and universities offer classes in a wide variety of activities at little or no cost. Often the entire family can participate in these classes—an added bonus to a total health improvement program. Also, the number of commercial fitness centers is rapidly growing, and most now employ trained staff who can properly prescribe exercise programs and help individuals get started.

Exercise and Rehabilitation of People with Diseases

Exercise has become a major component in **rehabilitation programs** for a number of diseases. Cardiopulmonary rehabilitation programs, which began in the 1950s, have become the most visible (see [chapter 21](#)). Tremendous advances in cardiopulmonary rehabilitation have led to the formation of a professional association, the American Association of Cardiovascular and Pulmonary Rehabilitation, and a professional research journal, *Journal of Cardiopulmonary Rehabilitation and Prevention*.

Exercise is also an important part of the rehabilitation of people with

- cancer;
- obesity;
- diabetes;
- renal disease;
- osteoporosis;
- arthritis, chronic fatigue syndrome, and fibromyalgia; and
- cystic fibrosis.

Most recently, emphasis on the use of exercise in the rehabilitation of transplant patients, including those with heart, liver, or kidney transplants, has increased because exercise helps alleviate some drug side effects and improves general health.

Although the specific physiological mechanisms explaining the benefits of exercise training for each of these diseases have not been clearly defined, exercise training carries many general health benefits that appear to improve the patient's prognosis. The manner in which exercise is used in the rehabilitation of people with disease is highly specific to the nature and extent of the disease. It is therefore beyond the scope of this chapter to present specific details for any disease, but many resources are now available that provide extensive detail about establishing exercise programs for people with particular diseases and the clinical values of these programs.¹

In Review

- An exercise session should begin with a low-intensity, dynamic warm-up of a type similar to the cardiorespiratory endurance exercise prescribed; functional and stretching exercises follow to prepare the cardiovascular, respiratory, and muscle systems to work more efficiently.
 - Cardiorespiratory endurance activities should be performed three to five times each week.
 - Each endurance session should be followed by cool-down and stretching to prevent blood pooling in the extremities and muscle soreness.
 - Flexibility exercise should be performed slowly, and this phase of the program might be best included immediately after the endurance component.
 - Resistance training should begin with a weight that is 40% to 60% of the person's 1RM. This is the proper weight if the person can lift it about 10 times; if he or she can lift it fewer than eight times, a lighter weight is required.
 - Recreational activities should be included in an exercise program for enjoyment and relaxation.
 - Exercise is a vital part of rehabilitation for most diseases. The type and details of the rehabilitation program depend on the patient, the specific disease involved, and its extent.
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IN CLOSING

In this chapter, we have seen that the medical community now regards physical activity as vital to maintaining good health and reducing the risk of disease. We looked at the importance and practicality of both a medical examination and an exercise ECG in screening previously sedentary adults before prescribing exercise. We discussed the components of an exercise prescription and methods of monitoring exercise intensity. Finally, we reviewed the components of an exercise program and the role of exercise in rehabilitating patients with disease.

Now that we have seen the importance of exercise in disease prevention, we will look more closely at physical activity as it relates to specific disease states. In the next chapter, we turn our attention to cardiovascular diseases.

KEY TERMS

Borg RPE scale
exercise electrocardiogram (ECG)
exercise prescription
graded exercise test (GXT)
heart rate reserve
metabolic equivalent (MET)
mode
predictive value of an abnormal exercise test
rating of perceived exertion (RPE)
rehabilitation programs
sensitivity
specificity
training heart rate (THR)

STUDY QUESTIONS

1. How active are adult Americans today?
2. Discuss the concepts of sensitivity and specificity of exercise testing and the predictive value of an abnormal test. Of what value is this information in the establishment of policy mandating who should be exercise tested?
3. How can we get our population to be more active? What levels of exercise do we need to promote to help people gain the health-related benefits associated with exercise?
4. What six factors must be considered in the exercise prescription? Which of these is the most important?

5. Discuss the concept of a minimal threshold for initiating physiological changes with exercise training as it relates to the exercise prescription.
6. Discuss the various ways of monitoring exercise intensity, and give the advantages and disadvantages of each.
7. Describe the components of a good exercise program and their importance in the total program.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

